



CSE623 Machine Learning

Weekly Report 8

Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis

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Date of Submission: 11-04-2026

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2026-2027 (Winter Semester)

Aim:

The primary aim of this week was to do the classical machine learning pipeline for pixel-wise semantic segmentation of wheat spikes. This involved expanding the feature extraction engine from 9, 19, and 25 specialized features, optimizing a Random Forest classifier with a strict confidence threshold and adaptive threshold, and implementing semantic contour fitting to precisely localize dense, overlapping wheat heads.

Introduction:

Proper identification and separation of wheat spikes in the agricultural domain are an essential part of phenotype measurement and automated yield prediction. Although deep learning techniques are widespread, our solution relies on a traditional machine learning framework based on the Random Forest algorithm along with manually engineered features. Thanks to the mathematical modeling of biological characteristics by means of color, texture, and frequency features, we can separate wheat from other objects in a computationally inexpensive way that deep neural networks cannot provide.

Work Completed:

For 10 features:

- **Multi-Channel Feature Extraction** Instead of just using RGB values, the model builds a highly-dimensional feature vector per each pixel that encapsulates color, texture, and structural characteristics.
Color Space Transformation: Transforming from RGB into CIE Lab to separate luminance from chrominance, as well as calculating vegetation indices such as ExG and NGRDI to discern living matter from soil.
Texture Extraction: Using Local Variance and Sobel filters to calculate surface roughness, which is much higher for wheat spikes compared to flat leaves.
Structural Filters (Gabor Filters): Applying Gabor filters to detect elongated linear structures. Used as a “leaf suppression” mechanism to identify the directional nature of wheat leaves.
- **Stratified Pixel Sampling and Model Training** Given the large amount of data imbalance between spike pixels and everything else, a Stratified Sampling Approach has been applied in order to mitigate it:
Sampling the ground-truth masks in a certain Spike-to-Leaf-to-Soil ratio, creating an equally weighted dataset.
Training of Random Forest Classifier with 200 trees, due to the non-parametric nature and resistance to overfitting of the algorithm used. Also, feature ranking is necessary for classical segmentation.

- Elongation Penalty: The probability map from the Random Forest Classifier is adjusted based on the Gabor filter response. Pixels showing strong "linearity" characteristics of leaves will be penalized.

Dynamic Threshold Search: As opposed to a fixed threshold value, the classifier searches for an optimum value in the range between 0.65 and 0.80 that achieves the maximum mean Intersection over Union (mIoU) on the validation set.

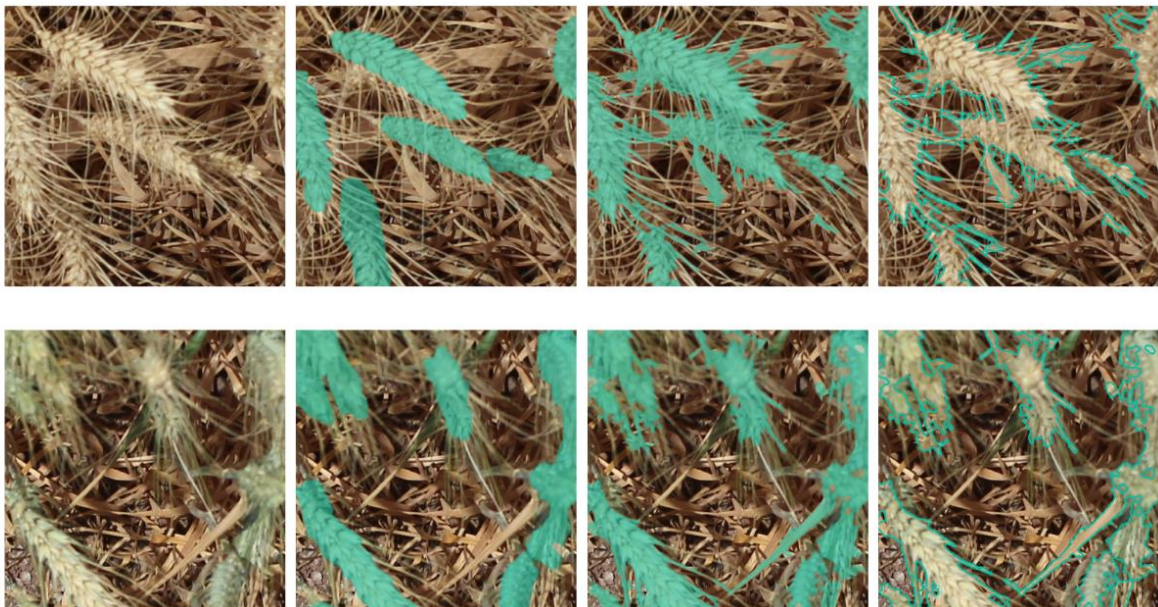
- Geometric/Morphological Operations: Successive application of Binary Open/Closing operations using circular structuring elements to fill holes and get rid of salt-and-pepper noise.

Filtering: Heuristic filtering based on the object's aspect ratio and solidity. Long and thin objects and objects with poor density are eliminated as spike artifacts.

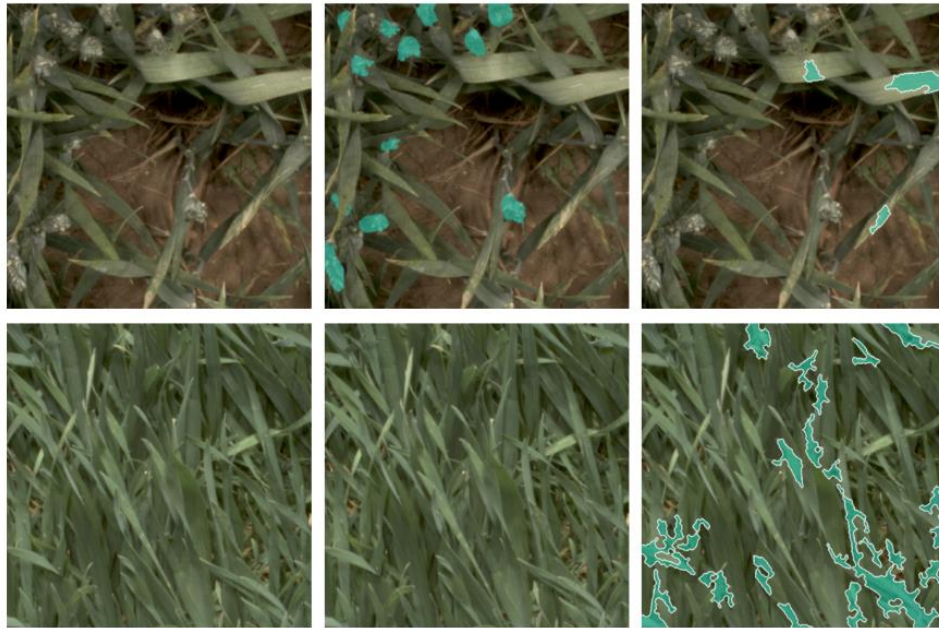
- Metrics mIoU: To calculate intersection over union (overlap) between predicted and ground truth spike pixels.
Pixel Accuracy: To assess the accuracy of the classifier across all three classes – Spike, Leaf, Soil.

RESULTS:

Training Samples:



Validation Samples:



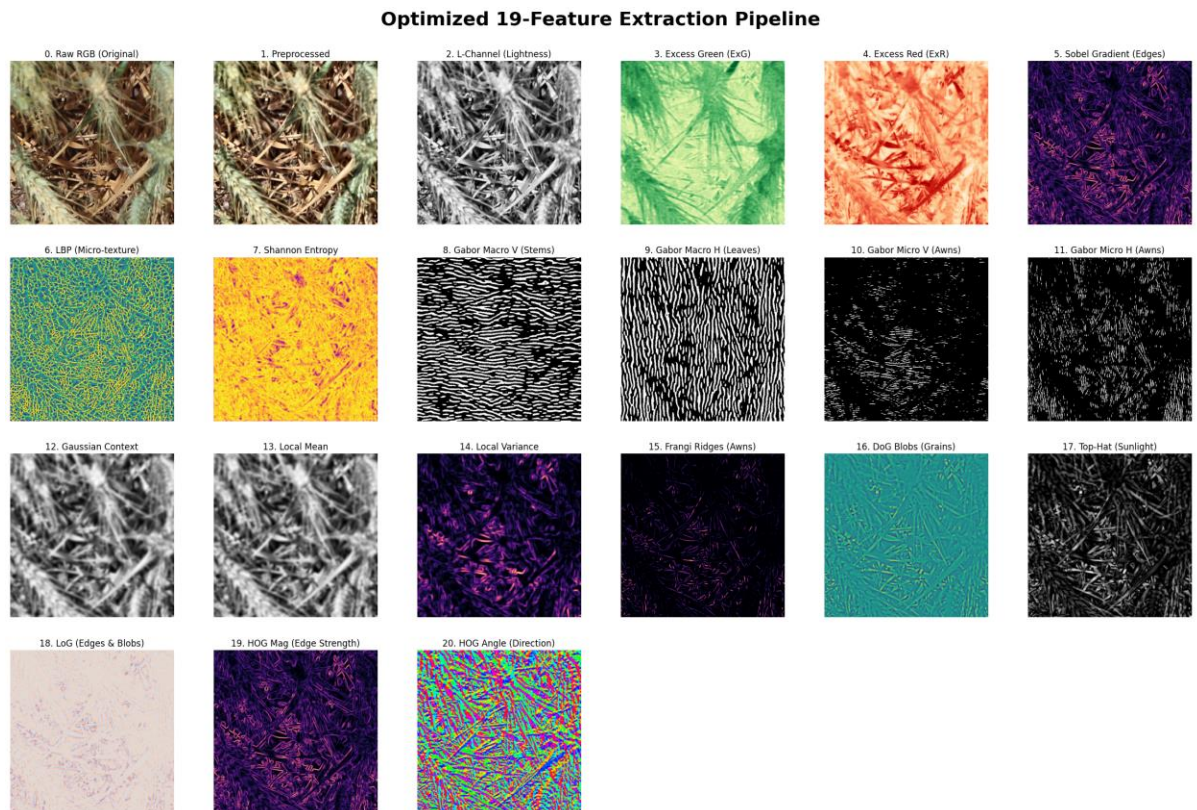
For 19 features:

This week we also did 19 feature extraction; we start by loading the datasets together with the associated ground truth. After that we preprocess the dataset for data quality improvement. The preprocessing methods employed include contrast enhancement using CLAHE and reduction of noise through Gaussian Blur. For each individual pixel in the image, a multidimensional array of features was obtained that captured different aspects of the image from different color spaces. The features captured are:

- Color features: LAB color channels, HSV components (hue and saturation), and normalized red and normalized green chromaticity.
- Biomass Features: Excess Green (ExG) and Excess Red (ExR) for the purpose of strictly discriminating plant materials from the soil background.
- Texture/statistical features: Local Binary Pattern (LBP), Shannon Entropy, local mean, and local variance matrices.
- Frequency/Directional features: Gabor filters of macro and micro scale for capturing orientation of stems and awns.
- Edge, Shape, and Morphological features: Sobel edge detection, Difference of Gaussians (DoG), Frangi filter for ridge detection, Top-Hat for illumination normalization, Laplacian of Gaussian (LoG), and Histogram of Oriented Gradients (HOG).

After extracting the feature vectors, the two-dimensional table of features was compiled for each image and store in .npz format.

RESULT:



Model Training:

For 19 features:

- The Random Forest classifier was trained using the 19 features. In order to address the class imbalance problem of the dataset, class balancing techniques were employed.
- Incremental training technique was used through the usage of the warm_start flag in order to increase the efficiency of the process of adding new trees. It is also possible to more effectively monitor the results of the training.
- Threshold of 0.50 was set for class prediction rather than using a default approach. The effectiveness of the model was assessed using metrics such as IoU, precision, recall, and F1 score. Moreover, a confusion matrix was calculated in order to examine the accuracy of the predictions.

- Training stopped early according to the validation IoU in order to prevent the overfitting. The trained model was stored.

RESULTS:

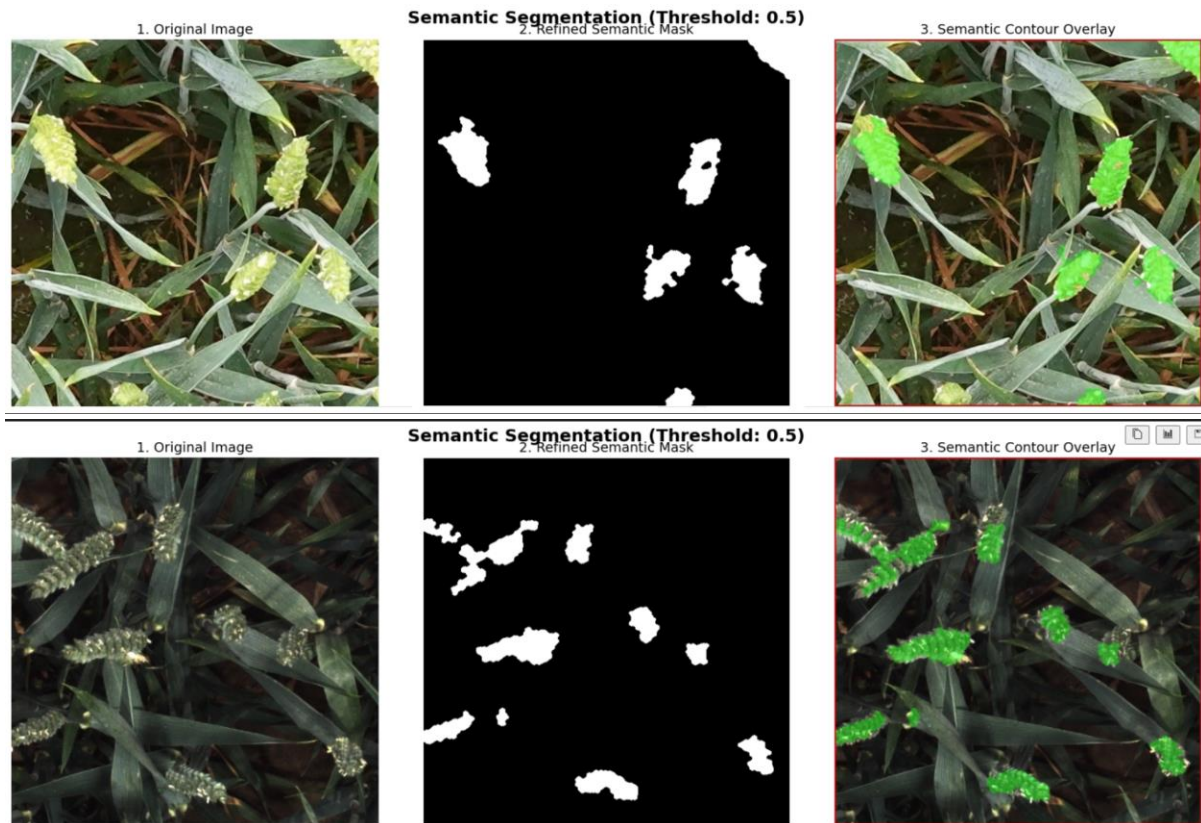
IoU:

```
Best Validation IoU: 0.3278
```

Model Prediction:



Semantic Segmentation:



For 25 features:

- For the advanced version, we train the Random Forest Classifier using 25 features that helped better represent the texture and structure and set the threshold to 0.65.
- The hyperparameters have been optimized to improve the generalization capability of the model and avoid overfitting. These include the tree maximum depth, minimum leaf sample size, and number of features per split, among others. For the purpose of prediction, a more conservative probability cutoff of 0.65 has been taken into consideration.
- As in the previous case, where there were 19 features, a dynamic training algorithm with the help of `warm_start` parameter was used. The performance of the model was being constantly tracked through metrics like IoU, precision, recall, and F1 score for both the train and validation datasets.
- When the improvement of the validation set performance stopped, training was interrupted.

RESULTS:

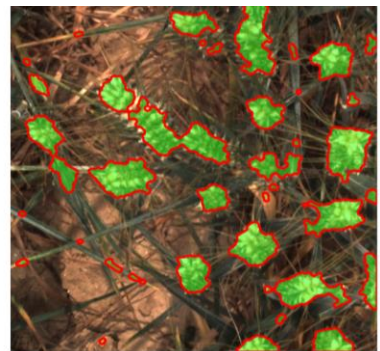
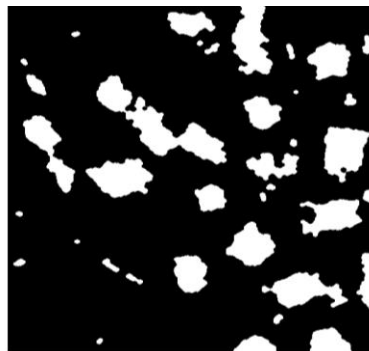
IoU:

```
Train IoU: 0.4739 | Validation IoU: 0.3457
```

Model Prediction:



Semantic Segmentation:



Next steps and goals:

Feature Space and Prediction of Yield

The first priority of future research will be to transform the present image segmentation pipeline into an entire Yield Prediction Framework, which will help turn the spike detection at the pixel level into valuable agricultural statistics. In order to make sure that the framework is going to function efficiently in practical applications, the next phase would involve extensive Feature Space Optimization.

As opposed to training and testing various machine learning models on a large set of features, the image segmentation pipeline will go through feature selection according to reliable feature importance criteria. Only those input values will be used that are known for their maximum Intersection over Union, namely features 9, 19, and 25. This procedure is expected to improve inference time while helping prevent overfitting. Ultimately, it will enable us to create an efficient yield prediction framework.

Conclusion:

In the project, a classic machine learning pipeline for segmentation of wheat spikes has been built by utilizing the algorithm of random forests along with engineered features. Performance has increased significantly with an increase in the number of features from 10 to 25; out of which, the 25 features model gave the highest accuracy.

References:

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